

Euler's Theorem

Module I

Euler's Totient function

- Euler's totient function, also known as phi-function $\phi(n)$, counts the number of integers between 1 and n inclusive, which are coprime to n .
$$\phi(n) = \text{count} (\{ 1 \leq x < n, \text{ where } x \text{ is coprime to } n \})$$
- Two numbers are coprime if their greatest common divisor equals 1 (1 is considered to be coprime to any number).

Examples:

$\phi(6) = \{1, 2, 3, 4, 5, 6\} = \text{coprime numbers with } 6 = \{1, 5\} = 2$

$\phi(9) = \{1, 2, 3, 4, 5, 6, 7, 8, 9\} = \text{coprime numbers with } 9 = \{1, 2, 4, 5, 7, 8\} = 6$

Properties of Euler's Totient function

- If p is a prime number, then $\gcd(p, q) = 1$ for all $1 \leq q < p$. Therefore we have $\phi(p): p-1$.

a) $\phi(5)=5-1=4$ [$\{1, 2, 3, 4, 5\} = \{1, 2, 3, 4\}$, here 5 is a prime number.]

b) $\phi(13)=13-1=12$

[$\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13\} = \{1, 2, 3, 4\}$, here 5 is a prime number.]

- If p is a prime number and $k \geq 1$, then there are exactly p^k / p numbers between 1 and p^k that are divisible by p . which gives us,

$$\phi(p^k) = p^k - p^{k-1}$$

- If a and b are relatively prime, then: $\phi(ab) = \phi(a) \cdot \phi(b)$

Properties of Euler's Totient function

- If p is a prime number and $k \geq 1$, then there are exactly p^k / p numbers between 1 and p^k that are divisible by p . *which gives us,*

$$\phi(p^k) = p^k - p^{k-1}$$

Example:

$$\phi(2^3) = 2^3 - 2^{3-1} = 8 - 4 = 4$$

$$\phi(8) = \{1, 2, 3, 4, 5, 6, 7, 8\} = \text{coprimes} = \{1, 3, 5, 7\} = 4$$

Properties of Euler's Totient function

- If a and b are relatively prime, then: $\phi(ab) = \phi(a) \cdot \phi(b)$ and a & b should be distinct.
- *Example1:* $\phi(15) = \phi(3) \cdot \phi(5) = 2 \cdot 4 = 8$.
- *Example2:* $\phi(49) = \phi(7) \cdot \phi(7)$. **X**
Instead, solve like this manner $\phi(7^2) = 7^2 - 7^{2-1} = 49 - 7 = 42$.
- *Example3:* $\phi(6) = \phi(2) \cdot \phi(3) = 1 \cdot 2 = 2$.

Exercises

- $\phi(20) = \phi(5) * \phi(2^2) = (5-1) * (2^2 - 2^{2-1}) = 4 * 2 = 8.$
- $\phi(21) = \phi(7) * \phi(3) = (7-1) * (3-1) = 6 * 2 = 12.$
- $\phi(19) = 19-1 = 18.$
- $\phi(45) = \phi(5) * \phi(3^2) = (5-1) * (3^2 - 3^{2-1}) = 4 * 6 = 24$

Euler's theorem

- *Euler's theorem states that if a and m are coprime positive integers then, (i.e. $\gcd(a, m) = 1$)*

$$a^{\phi(m)} \equiv 1 \pmod{m}$$

Example: $a=3$ and $m=10$. $\gcd(3,10)=1$.

$$a^{\phi(m)} \equiv 1 \pmod{m} \rightarrow 3^{\phi(10)} \equiv 1 \pmod{10} \rightarrow 3^4 \equiv 1 \pmod{10}$$

$$\phi(10) = \phi(5) * \phi(2) = 4 * 1 = 4$$

$$3^4 \equiv 1 \pmod{10} \rightarrow 3^4 \bmod 10 \equiv 1 \pmod{10} \rightarrow 81 \bmod 10 \equiv 1 \pmod{10}$$

$$1 \equiv 1$$

Euler's theorem

Example: $a=3$ and $m=10$. $\gcd(3,10)=1$.

$214^{512} \bmod 513 \rightarrow \gcd(214,513)=1$ coprimes.

$$214^{\phi(513)} \equiv 1 \bmod 513$$

$$\phi(513) = 513-1 = 512.$$

$$214^{512} \bmod 513 \equiv 1$$